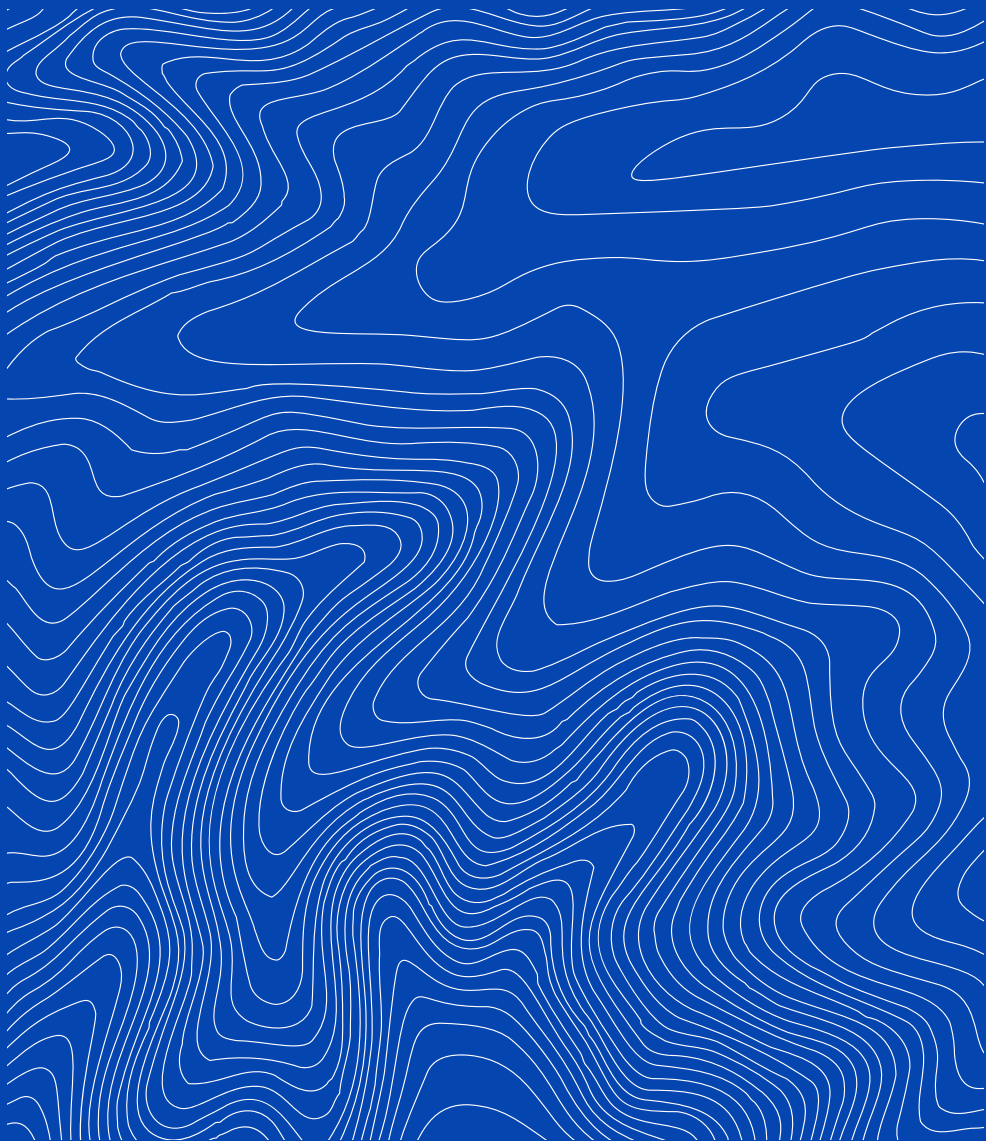


Cybersecurity Trends 2026



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Editorial

Anticipating risk in an increasingly complex environment

For years, cybersecurity has been approached from a primarily reactive perspective, focused on responding to incidents or complying with increasingly demanding regulatory frameworks. Today, that approach is insufficient. Business continuity, operational resilience and digital trust depend on the ability to anticipate risks, understand their systemic impact and make informed decisions in contexts of high uncertainty.

This report was created with that objective in mind. It is not just a matter of identifying technological trends, but of interpreting how the convergence of AI, OT environments, security platforms, digital sovereignty and new attack vectors is redefining the role of cybersecurity within organisations. In this new scenario, concepts such as Zero Trust, platformisation, technological geopoliticism and identity protection are no longer isolated initiatives but have become strategic pillars.

At S2GRUPO, we firmly believe that anticipating risk is not futurism, but responsibility. Committing to proprietary technology developed in Europe, advanced cyber intelligence capabilities and a deep understanding of critical environments is part of our commitment to digital sovereignty, resilience and the protection of strategic sectors. It is the way to deal with increasingly sophisticated attacks.

This report is an invitation to act before it is too late. To rethink models, question inherited assumptions and prepare for a scenario in which cybersecurity is no longer a technical requirement, but a decisive factor in competitiveness, trust and sustainability. Prioritising cybersecurity will enable us to tackle the most widespread threat facing businesses and society as a whole: disinformation inside and outside cyberspace.

José Rosell
CEO of S2GRUPO

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1 Introduction

By 2026, the attack surface will grow at a rate that exceeds the control capabilities of organisations, driven by the **proliferation of connected devices, the adoption of edge computing and the emergence of artificial intelligence agents capable of acting autonomously**. This scenario is blurring the traditional boundaries of security and multiplying the potential entry points for cybercriminals.

In this context, security models based exclusively on the perimeter are no longer sufficient. Architectures such as **Zero Trust** are becoming established as an **operational baseline**—especially in OT environments—where implicit trust is no longer a viable option. At the same time, to manage increasing complexity, organisations will increasingly turn to **automation, security platformisation and the use of autonomous systems**.

Attack
surface

Expanding
surface

Devices, Edge, and
autonomous AI multiply
entry points.

Perimeter
security

End of the
perimeter

Zero Trust replaces
implicit trust,
especially in OT.

Automation

Critical
automation

Technology speeds
things up, but human
judgment remains key.

However, speed and automation do not eliminate uncertainty. Interpreting weak signals, understanding the operational context and anticipating the real impact of an incident still require **specialised knowledge and human judgement**. The ability to correlate information, discriminate between what is relevant and what is incidental, and make informed decisions becomes a critical differentiator in defence.

At the same time, **geopolitical instability** is reshaping **technological decisions**. Technological sovereignty will become a strategic issue, directly linked to the resilience, operational continuity and competitiveness of organisations. Where systems are hosted, who controls the technology and under what jurisdiction data operates are now security decisions of the highest order.

This **2026 Trends Report** is based on analysis by LAB52, the cyber intelligence unit of S2GRUPO , and other company experts in security, intelligence and defence (teams such as industrial cybersecurity, R&D&I, regulatory compliance and behavioural security, among others), to offer an **overview** of the **risks and threats that will define this year**. Rather than anticipating futuristic scenarios, the aim is **to provide insights, analysis and alignment for informed decision-making**, helping CISOs and cybersecurity managers to prioritise actions in an environment marked by technological autonomy, global fragmentation and regulatory pressure.

In this scenario, **anticipating risk is not a matter of futurism, but of a deep understanding of the context**. And that is precisely where experience, cross-disciplinary knowledge and analytical skills make the difference.

Technological
sovereignty

Strategic
technology

Control and location of
systems affect resilience
and competitiveness.

Knowledge
and context

Decisive
intelligence

Interpreting weak
signals makes the
difference in defense.

2 Cybersecurity Outlook 2025 and what to expect in 2026

2.1

Executive summary

Data obtained over the past year suggests that **2026** will be marked by **more sophisticated threats and an expansion of the attack surface** driven by the **widespread adoption of digital technologies and the interconnection of supply chains**, which could also be affected by the **current international political situation, intensive use of the cloud and edge computing**.

The adoption of new technologies by threat actors could lead to a shift in cyberattacks, making them more persistent, better targeted and even quieter.

In this context, **artificial intelligence will play a key role**. Attackers could use it to automate highly personalised phishing campaigns, develop more adaptable malware and evade traditional security controls. In addition, companies will incorporate AI to test their defensive capabilities¹, including anomaly detection, automated incident response and risk prioritisation, accelerating reaction times and reducing operational impact.

²The strategic focus will shift from perimeter-based security to **Zero Trust models, continuous identity generation, and data protection**. Operational resilience, incident preparedness, and regulatory compliance will no longer be isolated initiatives but will be integrated into business governance, including greater involvement from senior management. Organisations will need to adopt a proactive, risk-based approach aligned with their strategic objectives to address the threat environment of 2026.

¹<https://www.bcg.com/press/18december2025-ai-cyber-threats-outpacing-defense-capabilities>

²<https://seraphicsecurity.com/learn/zero-trust/adopting-zero-trust-in-2025-a-practical-guide/>

2.2 Overview of the period

The use of AI and automation has facilitated the personalisation of phishing campaigns and the development of more effective³-based malware, which has also led companies to adapt to these threats by using AI-based tools, which will be discussed later in this report.

Ransomware, on the other hand, has established itself as the most widespread (up to 44% of data breaches in 2025 were related to ransomware) and profitable (on average, cybercriminals demanded £115,000 in ransom), evolving towards Ransomware-as-a-Service (RaaS) models and double or even triple extortion tactics with impacts on all types of organisations and sector⁴.

In addition to more technical factors, such as the exploitation of known or unknown vulnerabilities (zero days), geopolitics has had significant consequences in 2025 both inside and outside cyberspace, with disinformation and sabotage campaigns attributed to state actors⁵, which could continue to be a trend in 2026.

The human factor will continue to be a critical issue that organisations will need to address, as it is estimated that up to 88% of cyberattacks are directly or indirectly related to company employees⁶. Adopting zero trust and multi-factor authentication policies, as well as the importance of continuous training, will continue to be beneficial for organisations.

AI

enhances phishing personalization and malware development, redefining both attack and defense.

44%

of data breaches are related to ransomware.

88%

of cyberattacks directly or indirectly involve employees.

³ <https://www.kaspersky.es/about/press-releases/el-malware-que-imita-a-ias-como-chatgpt-se-disparan-un-115-en-2025>

⁴ <https://deepstrike.io/blog/cybersecurity-statistics-2025-threats-trends-challenges>

⁵ <https://www.eeas.europa.eu/sites/default/files/documents/2025/EEAS-3rd-ThreatReport-March-2025-05-Digital-HD.pdf>

⁶ <https://heydata.eu/en/magazine/human-weaknesses-in-cybersecurity-2025-risks-facts-and-solutions/>

2.3 Situation in Europe

In 2026, Europe will face a scenario of increasingly *sophisticated and complex cyber threats*⁷.

Actors posing a threat to Europe are carrying out increasingly rapid intrusions, developing new social engineering techniques and have resilient ecosystems at their disposal. This could be a consequence of the increasing digitisation of the economy, the increase in cross-border interconnection of services, the significant dependence on critical digital infrastructure, and the continuing geopolitical tensions at the global and regional levels within Europe. These trends will increase the risk that threat actors, especially ransomware and APT actors, will prioritise this type of digital infrastructure as a target⁸.

Ransomware actors will continue to pose one of the main threats to critical and strategic sectors such as energy, finance, manufacturing, automotive, transport, healthcare and public administration. In 2025, organisations located in Europe accounted for 22% of global victims of ransomware and extortion. It has also been detected that ransomware actors have significantly accelerated their attacks, even executing them in less than 24 hours from the initial breach⁹.



On the other hand, the adoption of artificial intelligence by hostile actors will enable them to automate and scale phishing campaigns, optimise their operations and develop social engineering techniques based on deepfakes, significantly increasing fraud, identity theft and extortion⁷.

Another relevant trend that will continue throughout 2026 will be attacks against the supply chain, highlighting the importance of indirect attack vectors through trusted partners of the potential victim⁸. Certain APT actors linked to nation states will carry out campaigns against the supply chain in order to exploit third-party dependencies and software supply chains to gain persistent access.

With regard to the type of activity carried out by hacktivist actors in Europe, beyond DDoS attacks and defacements, there has been an increase in malicious activity against industrial control systems (ICS), data breaches and unauthorised access against European organisations⁹.

Overall, Europe faces a changing cyber threat environment marked by the convergence of criminality, APT actors and hacktivism. The ongoing digitisation of services, dependence on critical infrastructure and geopolitical tensions could encourage hostile actors to carry out an increase in disruptive or exploitative operations against European organisations.

⁷ <https://www.crowdstrike.com/en-us/resources/reports/2025-european-threat-landscape-report/>

⁸ <https://www.crowdstrike.com/en-us/press-releases/crowdstrike-2025-european-threat-landscape-report-ransomware-hits-region-at-record-pace/>

⁹ <https://www.crowdstrike.com/en-us/press-releases/crowdstrike-2025-european-threat-landscape-report-ransomware-hits-region-at-record-pace/>

¹⁰ <https://www.isaca.org/about-us/newsroom/press-releases/2025/ai-driven-cyber-threats-are-the-biggest-concern-for-professionals-finds-new-isaca-research>

¹¹ <https://www.eye.security/blog/cyber-threat-landscape-outpacing-threat-actors-building-resilience>

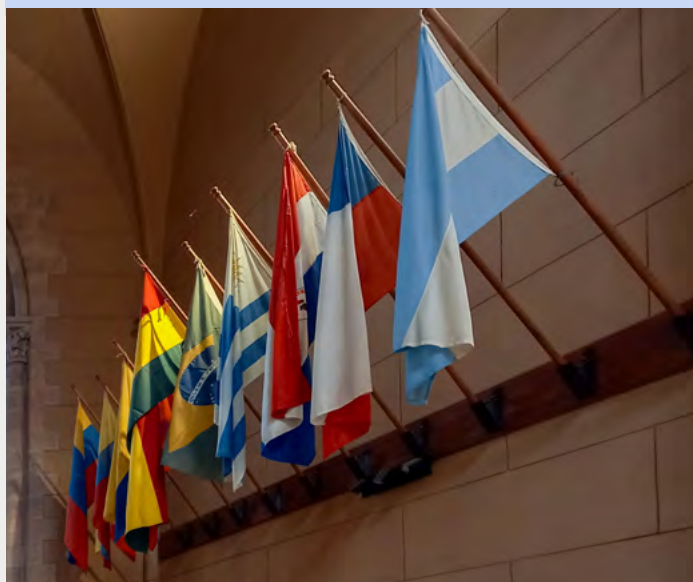
¹² <https://cyble.com/blog/hacktivist-attacks-critical-infrastructure-q3-2025/>

2.4 Situation in Latin America

In 2026, Latin America will face a more intense and complex cyber threat scenario, driven by accelerated digitalisation, the massive use of mobile services and persistent structural security gaps. The region will remain highly attractive to criminal actors, especially given the growing adoption of offensive artificial intelligence, which is capable of automating campaigns and increasing their speed and reach.

Ransomware will continue to be one of the main threats, with a particular focus on critical sectors such as energy, transport, telecommunications, manufacturing, healthcare, government and financial services.

<https://www.welivesecurity.com/es/informes/ia-ransomware-regulacion-tendencias-ciberseguridad-2026/>



Operational pressure and the integration of digital platforms make these sectors particularly vulnerable to disruption and extortion.

A notable trend will be the increase in mobile banking malware^{14, 15}, which will seek to steal credentials, intercept communications and manipulate transactions through fake applications and advanced device control techniques. This growth will be directly related to the region's dependence on mobile banking.

There will also be increased abuse of QR codes^{16, 17} and NFC¹⁸, taking advantage of their widespread use in payments and services. These vectors will enable phishing campaigns, installation of altered apps, and contactless fraud.

Organised cybercrime will continue to evolve and adopt more sophisticated tactics, particularly affecting public entities with limited infrastructure and legacy systems.

Faced with this scenario, organisations will need to strengthen strategies focused on identity management, segmentation, AI governance, mobile ecosystem protection and response automation. Resilience will be key to dealing with a threat environment that is more mobile, automated and focused on service disruption.

¹⁴<https://www.welivesecurity.com/es/malware/infostealers-presentes-latinoamerica-2025/>

¹⁵<https://cybersecuritynews.com/how-banking-trojan-grandoreiro-is-evolving-tactics-to-attack-victims-in-latam/>

¹⁶<https://www.infobae.com/colombia/2025/10/28/fraudes-con-qr-en-colombia-alcanzan-cifras-record-siga-estos-consejos-para-proteger-su-dinero/>

¹⁷<https://www.infobae.com/mexico/2025/12/14/qrishing-asi-es-la-nueva-ciberestafa-por-qr-que-simulan-multas-en-cdmx/>

¹⁸<https://cyble.com/blog/relaynfc-nfc-relay-malware-targeting-brazil/>

2.5 Harmful AI-driven campaigns:

Attacks and malicious campaigns driven by artificial intelligence were a clear trend in 2025 and are expected to intensify in 2026. ENISA warned in its latest Threat Landscape report about the use of artificial intelligence by foreign information manipulation and interference (FIMI) actors from Russia and China¹⁹.

In this context, the first massive AI-driven cyberattack was detected in November 2025, marking a turning point in the operations of threat actors. According to research²⁰, a Chinese-backed actor identified as CTG-1002 used Claude Code—a legitimate AI tool—to carry out a cyber espionage campaign against 30 companies worldwide, in which artificial intelligence autonomously executed approximately 90% of the actions. This case illustrates how AI has moved beyond being an experimental element to become a key operational enabler.



Based on this background, everything indicates that artificial intelligence will consolidate its position in 2026²¹ as a central attack vector. Its ability to automate and refine operations is powering techniques such as high-precision phishing, autonomous AI agents, detection mechanism evasion, and the use of deepfakes and highly convincing synthetic content for deception or the amplification of disinformation campaigns. Similarly, ESET notes that AI is also present in ransomware campaigns²², amplifying the impact of these threat actors. Furthermore, according to Google²³, financially motivated and state-backed threat actors can use AI-enabled malware in real-world environments, such as PROMPTFLUX and PROMPTSEAL. The tech giant also detected the sale of AI services to carry out illicit actions on underground forums.

The use of AI in cyberattacks brings with it a new conception of the concept of security, as it requires the inclusion of risk elements that were not previously included in protection models²⁴. If in 2025 IBM reported that 13% of organisations were victims of security breaches related to artificial intelligence models or applications, this trend will continue to rise in the coming years.

S2GRUPO has identified the main attacks and risk vectors against artificial intelligence systems²⁵. This table shows how attacks and dependencies are connected in the artificial intelligence life cycle:

¹⁹ https://www.enisa.europa.eu/sites/default/files/2025-11/ENISA%20Threat%20Landscape%202025_0.pdf

²⁰ <https://assets.anthropic.com/m/ec212e6566a0d47/original/Disrupting-the-first-reported-AI-orchestrated-cyber-espionage-campaign.pdf>

²¹ <https://www.welivesecurity.com/es/informes/ia-ransomware-regulacion-tendencias-ciberseguridad-2026/>

²² <https://www.welivesecurity.com/es/informes/ia-ransomware-regulacion-tendencias-ciberseguridad-2026/>

²³ <https://cloud.google.com/blog/topics/threat-intelligence/threat-actor-usage-of-ai-tools?linkId=17601224>

²⁴ <https://s2grupo.es/mas-alla-de-los-algoritmos-los-verdaderos-riesgos-de-la-ia/>

²⁵ <https://s2grupo.es/mas-alla-de-los-algoritmos-los-verdaderos-riesgos-de-la-ia/>

Attack	Affected phase	Main impact	Related to...
Prompt injection	- Input - Integrations	- Integrity - Confidentiality	Insecure output, insecure plugins, excessive agency
Insecure output management	- Inference - Operation	- Integrity - Physical security - Cost	Prompt injection, poisoning, bias, excessive dependency
Data poisoning	- Data - Training	- Model integrity	Insecure supply chain, insecure output
Data leakage	- Training - Inference	- Confidentiality - Compliance	Disclosure of confidential information, excessive dependency
Denial of service (DoS)	- Operation - Integrations	- Availability - Cost	Supply chain, excessive agency
Supply chain vulnerabilities	- Data - Training - Operation	- Integrity - Confidentiality - Availability	Poisoning, insecure output, insecure plugins
Disclosure of confidential information	- Integrations - Interference	- Confidentiality - Reputation	Data leakage, insecure output, excessive dependency
Insecure plugins	- Integrations - Operation	- Integrity - Confidentiality	Insecure output, excessive agency, supply chain

Attack	Affected phase	Main impact	Related to...
Excessive agency	- Operation - Business logic	- Systemic risk	Prompt injection, unsafe output, plugins, excessive dependency
Excessive dependency	- Governance - Operation	- Continuity - Sovereignty	Excessive agency, unsafe exit, disclosure
Theft of models	- Training - Inference	- P.I. - Evasion	Data leakage, poisoning, excessive dependency

2.6 Identified actors who have used AI:

LAB52 has detected that several cyber actors have used artificial intelligence. This number is expected to increase by 2026. Some of these prominent actors are:



2.6.1 Cybercrime:

2.6.1.1 FunkSec

Active since 2024, this RaaS has evolved from hacktivism, where it was aligned with Palestinian interests²⁶, to cybercrime. Governments, education, automotive, energy and IT are some of the targets of its global campaigns. FunkSec has used generative language models or similar tools such as Miniapps to promote the development of its malware.

FunkSec



2.6.1.2 GXC Team

Of Hispanic origin, its activities date back to 2023. Although its campaigns were initially focused on the Spanish online financial sector, it has also attacked countries such as the United States, the United Kingdom, Slovakia and Brazil. GXC Team has used LLM models to create more persuasive fake messages and pages without grammatical errors. Artificial intelligence has improved this threat actor's social engineering techniques, using technologies that mimic the voice of an alleged representative of a financial institution to make automated phone calls²⁷.

2.6.1.3 Indrik Spider

Of Russian origin, it has been active since 2024. Its targets include sectors such as finance, healthcare, technology, and manufacturing. This threat actor has been sanctioned by the United States²⁸ and has used artificial intelligence to optimise some of its operations. In 2023, a cybersecurity company responded to an incident involving the group related to BITWISE Spider's LockBit RED ransomware, in which Azure Key Vault credentials were leaked²⁹. The attackers used ChatGPT, search engines such as Google or Bing, and repositories such as GitHub to learn how to extract those credentials.

²⁶ <https://www.rapid7.com/blog/post/2025/06/03/from-ideology-to-financial-gain-exploring-the-convergence-from-hacktivism-to-cybercrime>

²⁷ <https://www.resecurity.com/blog/article/cybercriminals-implemented-artificial-intelligence-ai-for-invoice-fraud>

²⁸ <https://home.treasury.gov/news/press-releases/sm845>

²⁹ https://asantecloud.com/wp-content/uploads/2024/02/GlobalThreatReport2024_Asante.pdf

³⁰ <https://www.crowdstrike.com/adversaries/renaissance-spider/>

Renaissance Spider

2.6.1.4 Renaissance Spider

Also of Russian origin, it has been active since at least 2019. It has carried out influence and sabotage operations using false identities³⁰. Specialising in malspam campaigns, activity by this threat actor has been detected in Europe, Ukraine and Latin America.



2.6.2 APT groups:

2.6.2.1 APT28

Linked to the Russian GRU, its targets are governments, military institutions, think tanks, media outlets, and organisations related to NATO, the United States, Eastern Europe, and former Soviet bloc countries. This state actor used LLM to research satellite and radar technologies that could be used in military operations in Ukraine³¹. During these interactions, they requested information on satellite communication protocols, imaging radar systems, and other technical parameters. APT28 also used LLM to receive assistance with basic programming tasks.

APT28



2.6.2.2 Imperial Kitten

It is backed by the IRGC. Its targets include organisations related to sectors such as defence, technology, transport, telecommunications and logistics. This group uses language models to generate phishing emails or messages that influence its victims. It also uses these tools to solve problems that arise during its operations³².

2.6.2.3 Lazarus

Backed by North Korea's Reconnaissance General Bureau (RGB), this threat actor has been active since at least 2009. Lazarus mainly targets the finance, defence, energy, manufacturing, transport and critical infrastructure sectors. It has used LLM to enhance its targeted phishing campaigns and to research the profiles of its potential victims, as well as to search for information on vulnerabilities and solve technical problems³⁴.

³¹ <https://www.microsoft.com/en-us/security/blog/2024/02/14/staying-ahead-of-threat-actors-in-the-age-of-ai/>

³² <https://www.microsoft.com/en-us/security/blog/2024/02/14/staying-ahead-of-threat-actors-in-the-age-of-ai/>

³³ <https://www.crowdstrike.com/explore/2025-threat-hunt-report>

³⁴ <https://cloud.google.com/blog/topics/threat-intelligence/threat-actors-generative-ai-limited/>

Charming Kitten

2.6.2.4 Charming Kitten

It also has links to the IRGC and has been behind cyber espionage campaigns against Western governments, human rights organisations, journalists, academics and actors linked to national and nuclear security, as well as the aerospace sector, especially in the United States, Europe and the Middle East. Charming Kitten may have used AI to generate phishing attacks³³.



2.7 Notable OT attacks:

During the first quarter of 2025, nearly 22% of systems used in industrial environments globally recorded and blocked malicious activity, with a particular incidence in regions such as Africa and Southeast Asia. According to a report by Kaspersky , the industrial sectors most affected were building automation and the electrical energy sector, confirming the growing exposure of OT environments to both opportunistic and targeted threats.

Throughout 2025, several significant incidents were recorded that demonstrate the real impact of cyberattacks on industrial and essential service infrastructures. Among the most notable cases are:

A ransomware attack encrypted the servers of **Marposs**, an Italian manufacturer of measurement and control systems for the manufacturing industry. As a result, the company activated a temporary redundancy fund while operations were gradually restored³⁶.

A cyberattack against a dam in Norway allowed the attackers to open a floodgate and release water for several hours before the intrusion was detected³⁷.

In Denmark, a group of pro-Russian hackers accessed the control system of a water facility, causing an abnormal increase in pressure in the pipes that led to a physical breakdown of the system³⁸.

The **DarkSide** ransomware group compromised **Colonial Pipeline's** systems again. Although the impact was less than in the 2021 incident, control systems were encrypted and there were temporary disconnections in the operation of the energy infrastructure³⁹.

A cyberattack against **Schneider Electric** allowed malicious code to be inserted into software updates, also affecting SCADA/ICS systems deployed in production plants globally⁴⁰.

At the same time, in 2025, the Canadian Centre for Cybersecurity warned of an increase in malicious activities by hacktivist groups against ICS systems exposed to the Internet. Among the identified victims were a water treatment plant, an energy company and a farm. The most exposed components in these attacks were PLCs, RTUs, HMIs, SCADA systems, SIS, BMS and industrial IoT devices, reinforcing concerns about the attack surface associated with connected industrial environments.

Looking ahead to 2026, everything points to incidents against OT environments continuing to increase. Kaspersky anticipates a growth in attacks with the capacity to destabilise global logistics and high-tech supply chains. These campaigns could spread to less conventional infrastructure, such as intelligent transport systems, connected buildings and satellite communications. In addition, the adoption of AI agents and autonomous attack platforms will facilitate the execution of larger-scale industrial operations, with potential impacts that will go beyond digital disruption to reach the physical and operational plane.

³⁵ <https://me-en.kaspersky.com/about/press-releases/biometrics-and-building-automation-systems-were-the-most-attacked-operational-technology-sectors-at-the-beginning-of-2025>

³⁶ <https://ics-cert.kaspersky.com/publications/reports/2025/06/26/a-brief-overview-of-the-main-incidents-in-industrial-cybersecurity-q1-2025>

³⁷ <https://ics-cert.kaspersky.com/publications/reports/2025/10/09/a-brief-overview-of-the-main-incidents-in-industrial-cybersecurity-q2-2025/>

³⁸ <https://blog.denexus.io/resources/csb-2025-ot-security-lessons-industrial-control-systems>

³⁹ <https://otecosystem.com/global-ot-incident-roundup-quarterly-threat-brief-2025/>

⁴⁰ <https://blog.denexus.io/resources/csb-2025-ot-security-lessons-industrial-control-systems>

3 Trends



3.1 AI with agents: a new frontier in attack and defence

By 2026, the **accelerated adoption of artificial intelligence agents will transform the cybersecurity landscape**. Their ability to execute tasks autonomously, interact with third-party services and make chained decisions based on context will increase the operational efficiency of organisations, but will also significantly expand the capabilities of cybercriminals.

Thus, artificial intelligence will establish itself as a double-edged sword. According to the report *Cyber Threats and Trends 2024* by the National Cryptology Centre, AI is already being used to **create more credible phishing campaigns, generate malicious content using language models, and automate key phases of the attack cycle**, such as information gathering and vulnerability exploitation. Added to this is the rise of disinformation and the **use of deepfakes and⁴² tools** such as FraudGPT, WormGPT and DarkGPT, **designed to scale hostile operations against global targets**.

At the same time, **AI agents will be integrated on a massive scale into corporate workflows**. Attackers will exploit this same technology to **automate reconnaissance, lateral movement and data exfiltration processes**. The result will be an increase in more agile and personalised attacks, such as highly realistic vishing campaigns or business email compromise (BEC), driven not so much by the emergence of new techniques, but by the drastic reduction in costs, friction and technical barriers.

The adoption of AI in Europe will be conditioned by the entry into force of the European Artificial Intelligence Regulation (Regulation 2024/1689), which will be progressively enforced between 2026 and 2027, with some measures already in force from 2025. The framework introduces strict obligations, especially for High-Risk systems, and provides for significant financial penalties.

⁴¹ <https://www.ccn-cert.cni.es/es/informes/informes-ccn-cert-publicos/7274-ccn-cert-ia-04-24-ciberamenazas-y-tendencias-edicion-2024/file.html>

⁴² <https://cloud.google.com/blog/topics/threat-intelligence/adversarial-misuse-generative-ai>

⁴³ <https://www.consilium.europa.eu/es/policies/artificial-intelligence/>

At the same time, **data governance** will be consolidated as a **critical pillar**. The operational and regulatory impact of AI will make it essential to have strategies in place to ensure the quality, consistency and traceability of data throughout its entire lifecycle, reinforcing investment in data control and management mechanisms as the basis for secure and compliant adoption.

Given this scenario, **organisations** will need to **incorporate security, traceability and control measures directly into agent-driven systems**. The challenge will not only be to define what tasks AI can perform, but also **to set clear limits on its autonomy, ensure the observability of its decisions and prevent scenarios of excessive agency that could lead to systemic impacts**.

In this context, the **evolution of identity and access management (IAM)** will be a central element. **AI agents** should be treated as **distinct digital identities, with their own permissions and dynamic controls**. Google⁴⁴ anticipates identity management models based on continuous risk assessment, an approach that will reinforce already known principles—such as least privilege, just-in-time access, and controlled delegations—and will require a profound rethinking of cybersecurity control and accountability models.

⁴⁴ <https://services.google.com/fh/files/misc/cybersecurity-forecast-2026-en.pdf>

3.1.1 Recommendations: Governing AI agents with control and visibility

The incorporation of AI agents does not require reinventing cybersecurity, but rather extending already known principles to a new type of entity: autonomous systems that act, decide and integrate into critical processes. Preparing for 2026 will mean accepting that agents are not “users” or “traditional applications” and adjusting controls accordingly.

1

Prevention: defining limits from the design stage

Organisations must explicitly establish what an AI agent can and cannot do, avoiding deployments with excessive permissions. This will involve treating them as digital identities in their own right, applying principles of least privilege and just-in-time access, incorporating security by design into workflows, and defining clear thresholds of autonomy that determine when human intervention is necessary.

2

Detection: observing behaviours and decisions

With AI agents, detection will need to evolve beyond the analysis of isolated events. Many relevant deviations will be contextual, logically consistent but operationally incorrect, and derived from chained decisions. It will be necessary to monitor sequences of actions, correlate them with the agent's expected objective, and detect progressive drifts that would not trigger traditional alerts. In this scenario, agents will also be able to integrate into defence as operational assistants to security teams, supporting triage, signal correlation and initial analysis, always under human supervision and with full traceability.



Response: intervention and reversal capability

Coexisting with autonomous agents will require accepting the need to quickly stop or correct them. To do so, there must be clear *kill switch* or immediate deactivation mechanisms, complete traceability of decisions, and incident response plans that contemplate specific AI scenarios, including autonomy errors or misuse of integrations. Whenever possible, systems should be designed with reversal capabilities to avoid irreversible impacts.

The secure management of AI agents—from prevention and behaviour-based detection to response and reversal—requires continuous and specialised operational capabilities. In this context, **S2GRUPO CERT** provides 24/7 monitoring, triage, and incident response in environments where automation and the speed of attacks reduce reaction times.

Complementarily, **Advanced Cyber Intelligence** services make it possible to anticipate emerging tactics, identify the use of artificial intelligence by threat actors, and dynamically adjust detection models. Together, these capabilities make it possible to extend the classic principles of cybersecurity to a new type of digital actor, maintaining control, visibility, and responsiveness in a scenario marked by autonomy and automated decision-making.

3.2 Industrial IoT, *Edge Computing* and new devices increase the attack surface

The rapid adoption of ***edge computing***, the deployment of **5G and 6G networks** and the proliferation of IoT devices are significantly expanding the attack surface. This growth is also occurring in industrial contexts where security measures do not always reach the same level of maturity as in traditional data centres, increasing exposure to physical attacks, data interception and unauthorised access.

In industrial environments, this technological evolution has intensified the **convergence between IT and OT systems**. Automation, process digitalisation and integration with corporate systems, cloud services and analytics platforms have created a grey area in which it is increasingly difficult to distinguish which systems belong strictly to the industrial sphere and which to the corporate sphere. As a result, an incident in a system that is not initially perceived as industrial—for example, a corporate database, , or a management tool—can have a direct impact on production, causing plant shutdowns or disruptions to physical processes.

The **incorporation of Edge Computing** reinforces this trend. By moving part of the processing and control to distributed systems close to the machines, **efficiency and responsiveness are gained, but new risks are also introduced**. These edge nodes, often deployed in factories, logistics centres or a wide variety of infrastructures, may **lack consistent security controls, regular maintenance or adequate monitoring mechanisms, becoming pivot points for lateral movements within the network**.



At the same time, the **widespread use of IIoT (Industrial IoT) devices amplifies the problem**. Sensors, cameras, monitoring devices and connected equipment often have **long lifecycles, limited capabilities and uneven management**. Many rely on external suppliers, require remote access for maintenance or do not allow for simple firmware updates, making them particularly attractive targets for attackers. In this sense, **every device must be considered part of the supply chain**, with the associated risks that this implies.

The rapid expansion of these interconnected environments will allow both traditional and emerging attack vectors to be exploited, increasingly supported by artificial intelligence tools. Everything points to the fact that, by 2026, the **threat landscape associated with the IoT and Edge Computing will be marked by greater sophistication, automation and targeting of attacks**, with impacts that will go beyond information theft to operational disruption and even physical damage⁴⁵.

Estimates indicate that by 2026 there will be between 21 and 24 million connected devices in homes, businesses and critical sectors such as healthcare, transport and energy⁴⁶. This exponential growth will increase the risks arising from insecure configurations, obsolete firmware, lack of encryption and poor



credential management. **Critical infrastructures**⁴⁵ such as energy, water, transport and manufacturing will become **priority targets**, especially through attacks directed at control systems, sensors and distributed devices.

Among the most exploited vulnerabilities are the use of weak or default passwords, embedded credentials, insecure APIs, lack of encryption, and devices exposed to the internet without authentication. These weaknesses facilitate remote exploitation and massive compromise of equipment.

⁴⁶In this scenario, **IoT botnets** will continue to grow. Although they will continue to be used primarily for large-scale DDoS attacks, their integration into more complex networks—such as ORBs—will enable **more sophisticated offensive operations**. The incorporation of AI into these types of attacks will facilitate adaptive scanning, automated lateral movement, and advanced detection evasion techniques, further raising the level of risk for industrial and distributed environments.

⁴⁵<https://infinum.com/blog/cybersecurity-trends-2026>

⁴⁶<https://comparecheapssl.com/iot-security-statistics-threats-trends-safeguards-in-a-connected-world>

⁴⁷<https://www.cybersecuritydive.com/news/mobile-iot-attacks-surge-critical-infrastructure-zscaler/805008/>

⁴⁸<https://thehackernews.com/2025/10/experts-reports-sharp-increase-in.html>

3.2.1

Recommendations: Reduce the attack surface in industrial environments

1

Understand the environment to be protected:

No two industrial facilities are alike. Effectively reducing the attack surface requires a detailed understanding of production processes, the systems involved, and the actual dependencies between IT, OT, IoT, and Edge environments. Generic measures or those copied from other contexts are often ineffective and even counterproductive.

2

Have specialised industrial cybersecurity profiles:

They must be able to understand how production processes work, identify which systems are truly critical, and distinguish which interconnections are necessary to ensure business continuity. This knowledge allows for proper prioritisation, reduces unnecessary exposure, and avoids controls that do not add value or generate operational friction.

3

Work in coordination with plant teams:

Operations teams provide essential insight into the daily functioning of the facilities. Their participation allows for the identification of real risks, the validation of changes, and the implementation of security measures without introducing disruptions, operational risks, or impacts on physical security.

Reducing the attack surface is not just a matter of deciding what measures to apply, but of defining *how* and *where* to do so. The key is to **combine technical knowledge of cybersecurity with industrial experience and coordinated work with operational teams** to protect the activity without jeopardising its functioning.

Managing the risk associated with IoT, *Edge Computing* and IT/OT convergence requires continuous operational capabilities that combine technical visibility, access control and in-depth knowledge of industrial processes. **S2GRUPO** provides **industrial cybersecurity services** aimed at identifying assets, segmenting and ing environments, monitoring behaviour and managing vulnerabilities in complex and highly interconnected facilities.

This approach effectively reduces the attack surface associated with distributed devices and interdependent industrial systems, maintaining operational continuity, functional safety and control over environments where a technological incident can have a direct impact on production.



3.3 Zero Trust in 2026: From innovative strategy to survival standard⁴⁹

By 2026, the concept of the “network perimeter” will lose relevance in organisations’ protection strategies. The acceleration of hybrid environments, the consolidation of remote working and the massive integration of SaaS services will confirm an inescapable reality: **identity will become the new firewall** ^{50, 51, 52}.

In this context, Zero Trust architecture will cease to be perceived as an emerging trend and will become established as the **necessary operational baseline for any organisation that aspires to maintain the integrity of its data and ensure business continuity in the second half of this decade**. The traditional perimeter security model, based on the “castle and moat” approach, will give way **to architectures focused on identities, sessions, and dynamic access contexts**.

This change will profoundly redefine the allocation of budgets and resources. Investments will shift away from physical security appliances and th , designed to protect corporate headquarters or data centres, and will progressively move towards **identity and access management (IAM) solutions, SASE architectures and behavioural analytics tools capable of assessing risks in real time**. The focus will no longer be on securing “the cable” or the internal network, but on **protecting every session, every access and every interaction, regardless of its origin**.

⁴⁹ The projections in this chapter are based primarily on an analysis of the CISA Zero Trust Maturity Model implementation roadmaps, Gartner’s SASE adoption forecasts for 2025-2027, and the compliance requirements derived from the European NIS2 directive.

<https://www.microsoft.com/en-us/security/security-insider/threat-landscape/microsoft-digital-defense-report-2024>

<https://www.gartner.com/en/newsroom/press-releases/2023-01-23-gartner-predicts-10-percent-of-large-enterprises-will-have-a-mature-and-measurable-zero-trust-program-in-place-by-2026>

<https://www.sans.org/white-papers/sans-2024-state-ics-ot-cybersecurity>

<https://www.ccn.cni.es/es/normativa/directiva-nis2>

The adoption of Zero Trust will also consolidate its position as the **main enabler of hybrid and multi-cloud architectures**. By eliminating implicit trust based on physical location, organisations will be able to deploy critical workloads in public clouds with security levels comparable to those of on-premise environments. By 2026, Zero Trust will no longer be seen as a complement to cloud strategy, but as a **prerequisite for operating distributed infrastructures without taking on unacceptable risks**.

However, despite the maturity achieved by Zero Trust-related technologies, the transition will continue to present **significant obstacles**. From a technical standpoint, one of the main obstacles will continue to be **technological debt**. Many legacy applications will not support modern authentication and authorisation protocols, requiring complex adaptations to integrate them with identity proxies and advanced access control mechanisms, at a significant cost in terms of time and engineering resources.

Added to this challenge will be the **impact on the user experience**. When security policies are designed to be overly restrictive, friction will increase and productivity will be affected. The challenge in 2026 will be to strike a balance where **security is robust enough to mitigate real risks, but invisible enough not to disrupt daily workflow**.

Finally, the challenges will not only be technological, but also organisational and cultural. Zero Trust will require **close collaboration between traditionally separate teams**, such as Networks, Security, Infrastructure and Application Development. Resistance to sharing visibility, responsibilities and control will remain one of the most common points of failure in implementations. Overcoming these organisational silos will become as critical a factor as the technology itself in achieving a fully operational Zero Trust model.

⁵⁰ <https://www.checkpoint.com/es/cyber-hub/network-security/what-is-zero-trust/why-identity-is-the-new-perimeter/>

⁵¹ <https://www.beyondtrust.com/blog/entry/beyondtrust-cybersecurity-trend-predictions>

⁵² <https://abminfosys.com/20-emerging-cybersecurity-trends-to-watch-out-in-2026/>

3.3.1 Five steps towards Zero Trust maturity

Define the protection surface	1	The first step is not to deploy technology, but to identify which assets are truly critical to the business. Unlike the attack surface—which is broad and changing—the protection surface should focus on sensitive data, intellectual property, and high-value systems that require a priority level of control.
Map access and transaction flows	2	Achieving Zero Trust maturity requires understanding how information flows within the organisation. Identifying who accesses which resources, from where and under what conditions provides visibility into legitimate traffic and lays the foundation for access policies that do not compromise operations.
Designing the Zero Trust architecture	3	Organisations must rethink their security architectures from the inside out. Implementing security gateways and identity proxies between users and resources, regardless of their location, will eliminate implicit network-based trust and enable consistent controls in hybrid and multi-cloud environments.
Establish contextual access policies	4	Access control will evolve towards dynamic rules based on multiple variables, such as user identity, requested resource, device, time and purpose of access. This approach allows risks to be reduced in a granular manner, avoiding indiscriminate blocking and improving the user experience.
Implement continuous monitoring and adjustment	5	Zero Trust is based on constant observability. Inspecting and logging relevant activity, beyond perimeter traffic, allows for the detection of behavioural anomalies and the adaptation of policies in real time. Analytics will become a key component in anticipating incidents and responding with greater agility.

In 2026, Zero Trust will not be consolidated as a final destination, but as a **continuously evolving operational mindset**. Organisations that follow this roadmap **will** not only **reduce the impact of incidents and ransomware attacks**, but **will** also **gain resilience and flexibility**, decoupling security from physical infrastructure and preparing their business for future technological changes.

3.4 The great consolidation: From fragmentation to a unified cybersecurity ecosystem

In 2026, the trend towards consolidating security tools into a single platform will increase, as various reports have pointed out (Gartner Top Cybersecurity Trends or IBM Cost of a Data Breach Report)⁵³. After years of disorderly growth, complexity will consolidate as one of the main enemies of cybersecurity. The successive adoption of isolated solutions for endpoints, networks, email, the cloud and identity will have resulted in environments that are difficult to govern, where security teams operate a veritable technological "Tower of Babel".



In this context, **platformisation⁵⁴** will cease to be perceived as a trend driven by manufacturers and will become established as a **necessary operational response to alert fatigue, cost inefficiency and loss of visibility**. Experience will show that more tools do not equate to more security and that, in many cases, fragmentation increases risk rather than reducing it.

The legacy model based on multiple disconnected solutions will reveal its limitations. The **lack of interoperability between tools will continue to create blind spots**, where information is not correlated and context is lost. A threat detected in email may go unnoticed at the endpoint or in the cloud, creating gaps that attackers will exploit to move laterally. At the same time, the operational management of dozens of solutions—configuration, bastioning, patching, and maintenance—will become unsustainable, increasing the likelihood of human error and insecure configurations resulting from administrative overload.



Added to these problems will be **hidden costs**⁵⁵. Beyond licensing expenses, organisations will face a **steady increase in integration, training and operating costs**, as well as duplication of functionality between tools. By 2026, this fragmentation will become one of the main destroyers of return on investment in cybersecurity.

In response, the industry will accelerate its shift towards **unified security platforms**. Organisations will move from purchasing individual tools to operating integrated ecosystems—such as SIEM, CNAPP or SASE—capable of centralising telemetry and providing a consistent view of risk. In these environments, the detection of an anomaly at a specific point can automatically trigger coordinated actions in other domains, from the endpoint to the cloud, all managed from a single control layer.

This approach will reduce redundancy, simplify management and restore a holistic view of the security posture. The **platform will act as a central point of reference**, something that the sum of isolated tools has failed to deliver consistently.

However, technological consolidation will not eliminate the need for human talent; on the contrary, it will redefine its role. Greater automation capabilities, driven by artificial intelligence, will free analysts from repetitive, low-value tasks, but **will make the role of the expert even more relevant**. By 2026, cybersecurity teams will tend towards hybrid models, where **advanced automation is combined with professionals capable of interpreting complex signals, validating decisions made by autonomous systems and acting strategically in unforeseen situations**.

In this scenario, analysts will cease to be console operators and take on a role closer to that of threat hunters and risk managers, supported by platforms that filter out noise and manage known information at machine speed.

⁵³ <https://digitalinside.es/la-sobrecarga-de-herramientas-de-ciberseguridad-complica-la-proteccion-empresarial/>

⁵³ <https://www.paloaltonetworks.com/cyberpedia/what-is-cybersecurity-platformization>

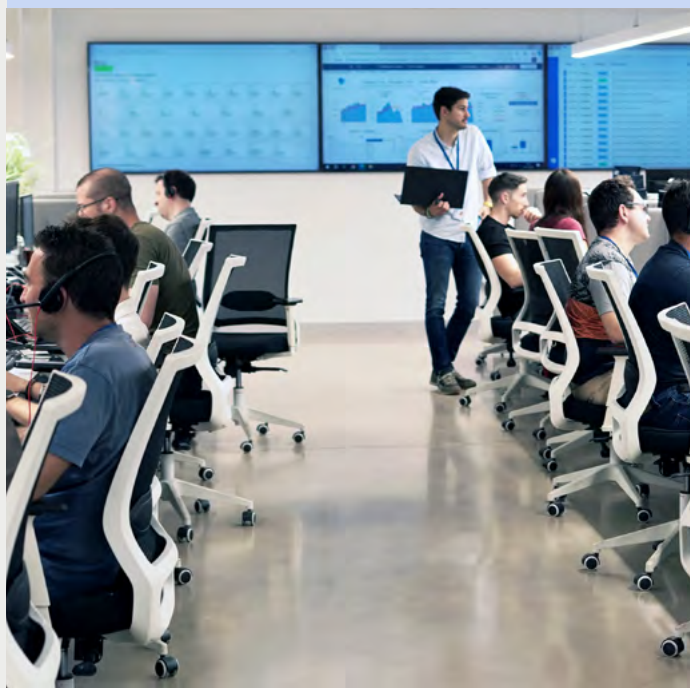
⁵⁴ <https://www.paloaltonetworks.com/cyberpedia/what-is-cybersecurity-platformization>

⁵⁵ <https://cdn.table.media/assets/wp-content/uploads/2024/07/30132828/Cost-of-a-Data-Breach-Report-2024.pdf>

Platformisation **will also transform the role of managed security service providers**. In an environment dominated by consolidated platforms, the value of MSSPs will no longer lie in the mere monitoring of alerts, but in their ability to manage complexity. **Technology agnosticism will become a key factor**: the MSSP of 2026 will stand out for its ability to operate, integrate and orchestrate different leading platforms, regardless of the manufacturer. Added to this will be cross-functional knowledge that will provide the layer of expert human intelligence that many organisations will not be able to maintain internally.

The lesson that will be consolidated by 2026 will be clear: complexity is the attacker's refuge. Simplification through platformisation **will reduce the management surface area, optimise costs and, above all, give control back to the defenders**. The goal will no longer be to have more tools, but to **have a unified and coherent view of the real state of security**.

The SOC as the nerve centre: Operating complexity



In this new paradigm, the Security Operations Centre (SOC) is no longer a passive monitoring space but becomes the operational core of defence. A unified platform only adds value if there is a SOC capable of interpreting correlated telemetry and orchestrating incident response; otherwise, consolidation translates into complexity and costly “noise”. The SOC’s mission is no longer to integrate tools, but to act with complete visibility and contextual judgement.

The main operational advantage lies in the shift from isolated alerts to contextualised incidents. By operating on consolidated platforms, analysts have access to complete attack narratives, significantly reducing detection and response times (MTTD and MTTR) and focusing efforts on neutralising real threats, rather than manually correlating signals.

This evolution also redefines the talent profile. Automation absorbs repetitive tasks, while the SOC of 2026 demands specialised profiles capable of cross-cutting analysis of identity, cloud and endpoint. The platform does not simplify the work: it elevates it, shifting the focus to strategic analysis and decision-making.

In this context, **S2GRUPO** offers a **Managed Security** service geared towards business resilience. By centralising operations and eliminating silos, the SOC can immediately execute global containment actions, a critical capability in a scenario where attacks, such as ransomware, are evolving at ever-increasing speeds.

3.5 Technological sovereignty: from global dependence to “geopatriación”

By 2026, geopolitical instability and regulatory tightening will be leading organisations to structurally rethink their dependence on global technology providers. This rethinking will no longer be limited to the cloud, but will encompass software, digital infrastructures, hardware and entire technology supply chains. In this context, **technological geopatriation**⁵⁶ will become established: **the relocation of workloads, services and strategic components to sovereign, local or regional environments with the aim of reducing geopolitical risks, strengthening regulatory compliance and regaining operational control.**

This movement will go beyond simple repatriation from hyperscale clouds. It will include supplier diversification (⁵⁷), the adoption of sovereign infrastructures and hybrid models with greater autonomy, and the reduction of critical dependencies that could become single points of failure. The goal will not be isolation, but resilience in an increasingly volatile international environment.

Tensions between the United States and China will act as one of the main catalysts for this trend. Restrictions on the export of semiconductors and critical technologies, together with cross-sanctions, are fragmenting the global technology ecosystem⁵⁸ and forcing companies and countries to redefine their technological alliances. This polarisation will increase costs, add operational complexity and hinder the global adoption and deployment of key technologies.

⁵⁹From a cybersecurity perspective, technological dependence on a single bloc or supplier will become a strategic risk. Hardware components, software or digital services from a single source

⁵⁶<https://emt.gartnerweb.com/ngw/globalassets/en/chief-information-officer/documents/2026-top-tech-trends.pdf>

⁵⁷https://www.europarl.europa.eu/doceo/document/A-10-2025-0107_EN.html

⁵⁸<https://modern.diplomacy.eu/2025/05/10/us-china-technology-competition-a-new-cold-war-threat-to-global-stability>

⁵⁹<https://www.uscisguide.com/information/u-s-technological-and-cybersecurity-concerns-in-china>

could become vectors for political pressure, interference or exposure to unilateral decisions, directly affecting business continuity.

For the European Union, technological geopolitism will take on a clearly strategic character. Although the region will continue to show a high dependence on non-European suppliers—especially in cloud infrastructure and advanced semiconductors—initiatives aimed at strengthening digital sovereignty will intensify in 2026. The deployment of regulatory frameworks, the promotion of critical technologies and increased investment in cybersecurity and artificial intelligence will seek to ensure jurisdictional, operational and political control of the most sensitive technological assets. While this strategy will not allow for full autonomy in the short term, it will reduce exposure to geopolitical risks and critical dependencies.

In Latin America, geopolitism will manifest itself more gradually and heterogeneously, driven mainly by national regulatory initiatives and the need to strengthen resilience to external disruptions. In this context, Chile and Brazil stand out, where technological dependence on large foreign corporations (mainly American) and regulatory weakness in some countries in the region are a reality. The development of regional technological alternatives is therefore expected in an attempt to reduce



dependence on large global suppliers, as well as the promotion of more regulations by some countries currently lagging behind in areas such as privacy, AI or information hosting requirements for strategic sectors (energy, health, finance, government, industry). Specifically, Chile has positioned itself as one of the regional leaders following the enactment of the Framework Law on Cybersecurity and Critical Information Infrastructure (Law 21.663), which establishes obligations for operators of essential services and strategic sectors. Such regulatory frameworks not only strengthen the protection of critical infrastructure, but also promote the development of local capabilities and the progressive reduction of external technological dependencies, laying the foundations for digital sovereignty adapted to the reality of the region.

In the field of artificial intelligence, this logic of geopoliticism will also begin to materialise at the regional level. The launch of Latam GPT marked a turning point for Latin America, demonstrating the viability of locally trained AI models aligned with the cultural, linguistic and regulatory contexts of the region. Looking ahead to 2026, this emerging local AI industry—still in its infancy but increasingly coordinated—is expected to make progressive advances as part of a broader strategy aimed at reducing external technological dependencies and strengthening digital sovereignty with a Latin American identity.



Overall, technological geopoliticism will consolidate in 2026 as a pragmatic response to a fragmented global environment. Rather than a retreat, it will represent a reconfiguration of dependencies aimed at gaining sovereignty, resilience and control in a cyberspace increasingly conditioned by geopolitics.

In this sense, one of S2GRUPO's strategic axes is digital sovereignty and the development of proprietary technologies, designed and operated from Europe, aligned with regulatory frameworks and the requirements of critical sectors. This approach is embodied in solutions such as MicroClaudia, Claudia and Carmen, designed to provide advanced detection, analysis and response capabilities without relying on external technologies subject to geopolitical decisions, sanctions or access restrictions.

In addition, S2GRUPO's cyber intelligence and threat analysis services enable organisations to anticipate risks arising from geopolitical tensions, assess critical technological dependencies and adapt their security strategies to a fragmented international environment. This vision is key to identifying exposures in the digital supply chain, strategic suppliers and sensitive technological assets.

4 Other trends

Other issues have also begun to gain relevance which, although not yet perceived as priorities for many organisations, will be **decisive in the coming years. Post-quantum cryptography** and **identity protection** against artificial intelligence-driven attacks fall into this category.

At **S2GRUPO**, we address these areas as part of our **anticipation strategy**: identifying risks before they materialise, preparing organisations for future scenarios and laying the technical, operational and governance foundations necessary to tackle them with an advantage. These emerging trends define not only the future of cybersecurity, but also the ability of organisations to anticipate risk and respond with resilience.



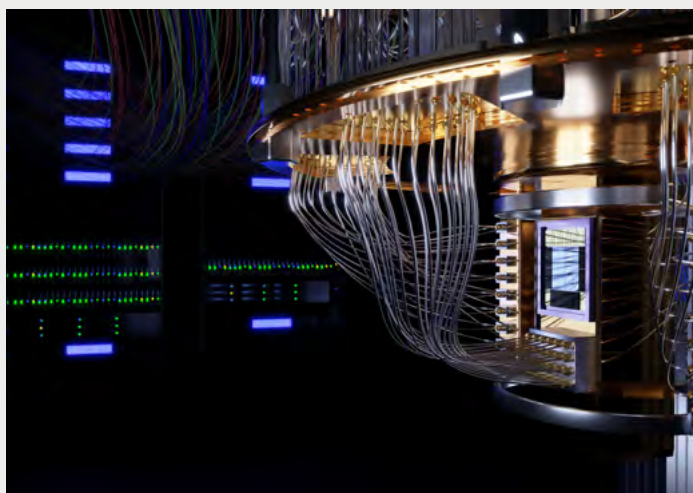
4.1

The era of post-quantum cryptography

Quantum computing will radically transform the cybersecurity landscape by making it possible to break current cryptographic systems. Problems that today would require trillions of years of computation on classical computers could be solved in days or hours using quantum algorithms, exposing information that is currently considered secure.

Although there is no specific date for *Q-Day*—the moment when quantum computing will be able to compromise traditional cryptography—the risk is already tangible. Data with a long shelf life, such as medical records, financial information, or public administration records, could be intercepted today and decrypted in the future.

In this context, the **adoption of post-quantum cryptography (PQC) is becoming a strategic necessity**. In 2026, preparing for quantum computing will not be a matter of anticipating when Q-Day will arrive, but of acting in advance by **identifying cryptographic assets, analysing quantum risks and defining a roadmap for the progressive adoption of quantum-resistant algorithms**. Waiting until the threat is imminent will mean arriving too late.



4.2 Identity protection in the age of AI

By 2026, **identity** will be established as the new perimeter. The evolution of social engineering, driven by artificial intelligence, will be shifting risk from technological infrastructure to human behaviour. Attackers will no longer need to exploit complex technical vulnerabilities: they will seek to manipulate legitimate decisions within real processes, taking advantage of contexts of trust, urgency or authority.

The use of AI will enable the scaling and refinement of techniques such as **business email compromise, vishing and impersonation using voice, image or video deepfakes**, eroding traditional human verification mechanisms. At the same time, the ungoverned use of AI tools by employees will introduce new risks of exposure of sensitive information and regulatory compliance.

In this context, identity protection will require going beyond classic technical controls. Organisations will need to incorporate **behaviour-based approaches, combining technical safeguards with data-driven awareness** to continuously identify, measure and reduce human risk. Identity will cease to be a static attribute and become a dynamic element that must be protected, monitored and governed in real time.

5 Key points for the CISO



1

Govern autonomy before scaling it.

La adopción de agentes de IA exige definir límites claros, trazabilidad de decisiones y capacidad de intervención; sin control, la automatización amplifica errores y riesgos sistémicos.

2

Assume that the attack surface is distributed and persistent.

Every device, sensor, or edge node must be managed as part of the supply chain, with visibility, segmentation, and lifecycle control from design onwards.

3

Treat Zero Trust as an operating principle, not a one-off project.

Identity—human and non-human—becomes the new perimeter and must be governed dynamically in IT, OT, cloud, and SaaS.

4

Reduce complexity to gain control.

The priority is no longer to add tools, but to consolidate visibility and context to move from isolated alerts to actionable ones with a real impact on response.

5

Incorporate geopolitics into the risk map.

Where systems are hosted, who controls the technology, and under what jurisdiction the data operates are strategic decisions for cybersecurity and business continuity.

6 About S2GRUPO

We are the leading Spanish company in cybersecurity, intelligence and defence. We not only protect infrastructures, we transform cybersecurity into a strategic asset so that your organisation can grow, innovate and prosper in an interconnected and unpredictable world.

Independent Pure Player

We offer independent services and customised solutions aligned with customer needs.

We specialise in high-impact sectors such as public administration, defence, industry, energy and healthcare, providing solutions tailored to the specific needs of each.

Industry expertise

We have cross-functional capabilities and extensive experience as a true end-to-end partner in IT and OT. We lead the convergence of Information Technology (IT) and Operational Technology (OT), offering integrated solutions for the protection of critical infrastructures.

Comprehensive solutions

Operational excellence throughout the security lifecycle

From early threat detection to advanced response capabilities, we guarantee exceptional results throughout the entire security cycle, even in the most complex and high-risk environments.

We offer reliable, autonomous solutions that comply with the regulatory standards of the territories in which we operate, thus guaranteeing our customers' technological sovereignty.

Technological independence and sovereignty

Does your organisation need more than just a supplier?

We offer reliable, autonomous solutions that comply with the regulatory standards of the territories in which we operate, thus guaranteeing our customers' technological sovereignty.

**Book a personalised consultation
with our experts**

Logbook of the invisible

A comprehensive analysis of one of the most advanced espionage artefacts attributed to APT29, characterised by a very high level of technical sophistication. Discover how the malware is structured, what mechanisms it uses to go undetected and what tactics were used to decode it.

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NIS2 Directive

We present the key points that every company needs to know in order to comply with regulations, protect its digital assets, avoid penalties and move forward in building its cybersecurity strategy.

**Download
Whitepaper NIS2 >**

Beyond algorithms: the real risks of AI

We take an in-depth look at how AI risks manifest themselves on three levels—technical, operational and human—who exploits them, and what strategies can be used to mitigate them. A clear and practical guide for CISOs seeking to anticipate the new threat landscape and protect innovation without stifling its potential.

**Download Risks of
IA report >**

Ransomware Outlook 2025

We analyse the most significant ransomware events of 2025 and what organisations should do to protect themselves against this threat.

**Download the report - Ransomware
Overview 2025 >**

Appendix

Details of attacks perpetrated with artificial intelligence:

p

Prompt injection.

The attack consists of manipulating the inputs processed by the model, such as prompts, documents or external data, to modify its behaviour, cause unexpected actions or obtain confidential information by exploiting an injection vulnerability in which modified or false data is interpreted as instructions. Because these actions affect the logic of the model, they can go unnoticed by defence mechanisms. This attack is direct when the user's prompt is altered or indirect when malicious instructions are hidden in external content.

i

Insecure output management.

When AI outputs are used directly in business operations, any erroneous or manipulated results can have serious consequences. This risk may reflect the presence of previous vulnerabilities, such as prompt injections, data contamination, or hidden biases in model training.

d

Data poisoning.

The model's behaviour can be altered by intentionally introducing malicious data during the training or updating phases, which can affect its performance, distort results, or incorporate harmful biases that compromise its integrity and reliability. This type of compromise can originate from compromised external sources, data providers, or automated integrations with untrustworthy APIs.

d **Data leakage.**

The model may reveal sensitive information acquired during its training or stored in its internal memories. This leakage can occur through malicious queries, model extraction attacks, or configuration errors that allow access to personal, confidential, or strategic data.

d **Denial of service.**

The attacker can overload an AI model with a massive volume of requests, complex queries, or manipulated data, which can degrade its performance, exhaust resources, or artificially inflate execution costs in cloud environments. Although this is a traditional attack, in the context of AI it takes on greater relevance because these models require high memory and GPU resources and are often integrated with external services or public APIs that expand their exposure surface.

S **Supply chain vulnerabilities.**

The use of libraries, frameworks, pre-trained models, and external services can introduce inherited vulnerabilities that compromise the entire AI pipeline, even if the model itself remains intact. A failure in a common dependency can allow remote code execution or data manipulation, affecting the systems where AI is integrated. This risk combines technical exposure due to vulnerable or outdated components with operational dependence on external vendors and repositories.

d **Disclosure of confidential information.**

AI models can leak sensitive or private information as a result of malicious queries, incorrect access configurations, or lack of validation on their outputs. This can include personal data, corporate information, internal parameters, or fragments of the training set. The risk is especially high in environments where AI will interact directly with external users or other information systems without supervision.

m

Manipulation of insecure plugins.

External integrations and plugins extend the capabilities of the system and open up new attack vectors if installed without control or validation. In the case of AI, the impact of any manipulation is amplified by allowing the model to interact directly with databases or communicate with external services. In this way, an attacker could exploit a vulnerable or poorly designed plugin to inject instructions, alter data, or redirect results, compromising the integrity of the system and the information it processes.

e

Excessive agency.

This risk arises when an AI agent has the autonomy to perform critical actions without proper human supervision. Lack of control can transform errors or manipulations into direct and immediate impacts on business processes. Excessive agency amplifies and links existing risks.

e

Excessive dependency.

This occurs when an organisation relies too heavily on AI models or the ecosystem of services that support them. It involves basing all operations on a network of external models, APIs and frameworks, which increases exposure and complicates governance.

m

Model theft.

Model theft or extraction occurs when an attacker manages to reproduce the behaviour of an AI model through systematic queries or directly accesses its parameters and weights through social engineering techniques. This compromises intellectual property and makes attacks easier. API exposure, unauthenticated integrations and open code repositories are factors related to model theft.





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